
Voodoo Dough Boys

The Calculatorinator

Test Case

Version 1.2

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

Revision History

Date	Version	Description	Author
30/04/26	1.0	Add to sections 1 and the test case table	Joseph Ubanda Tram Dinh
04/05/26	1.1	Revised test case table	Maddy Dang
04/05/26	1.1	Sort cases by new ID types and added new test cases	Joseph Ubanda
05/05/26	1.2	Add to the test case table	Micah Lehman
05/07/26	1.2	Add to the test case table	Maddy Dang
05/07/26	1.2	Fill out sections 3,4 and 5	Vu Nguyen Joseph Ubanda

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

Table of Contents

1. Purpose	4
2. Test case identifier	4
3. Test item	4
4. Input specifications	4
5. Output specifications	4
6. Test Cast Table	4-9

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

Test Case

1. Purpose

This Test Case Specification document for the Calculatorinator defines a test case for an item that should be tested. The test cases include valid expressions using addition, subtraction, multiplication, division, modulo, exponent, and unary operators. Valid expressions can also use a combination of operators with parentheses. There are also invalid expression test cases for different possible errors like division by zero, missing values or operators, and syntax errors.

2. Test case identifier

Specify the unique identifier assigned to this test case.

ADD: Addition operator test case
SUB: Subtraction operator test case
MULT: Multiplication operator test case
DIV: Division operator test case
MOD: Modulo operator test case
EXP: Exponent operator test case
UNA: Unary operator test case
MIX: Mixed operators test case
FLOAT: An expression with float values
ERR: test cases that cause errors
Q: Test cases quitting the operations

3. Test item

The description part of the test case table explains what function of the program that a certain test case is testing. Each will focus on a certain goal or feature that it will test, and the code is organized within the table depending on what feature is being tested.

4. Input specifications

The test cases of this program will be defined as user inputs. Each input will be the expression or value that the user enters into the program on the command line while the program is running. Test inputs will consist of numbers, letters, and operations to test a variety of program functions to ensure that the program runs as expected.

5. Output specifications

The three rightmost columns list the expected output, actual output, and results. The expected output is the expected result of the input expression. The actual output is the evaluated result the program outputted. The result determined if the expected and actual outputs match and resulted in the correct answer for the expression.

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

ID	Description	Input	Expected Output	Actual Output	Result
Q01	Quitting	q	Thank you for using the Calculatorinator!	Thank you for using the Calculatorinator!	Pass
Q01	Quitting	Q	Thank you for using the Calculatorinator!	Thank you for using the Calculatorinator!	Pass
ADD01	Addition	3 + 4	7	7	Pass
ADD02	Complex Addition with Extraneous Parentheses	$((2 + 3)) + ((1 + 2))$	8	8	Pass
ADD03	Unary Negation and Addition in Parentheses	$-(+1) + (+2)$	1	1	Pass
ADD04	Large Integer Addition	939292993 + 1111111	9.40404e+08	9.40404e+08	Pass
SUB01	Subtraction	8 - 5 - 2	1	1	Pass
SUB02	Subtraction with parentheses	9 - (10 - 6)	5	5	Pass
SUB03	Unary Negation and Subtraction	--3-5	-2	-2	Pass
SUB04	Large Integer Subtraction	999999999 - 999999199	800	800	Pass
SUB05	Negative Result from Subtraction	3-7	-4	-4	Pass
MULT01	Multiplication	10 * 2	20	20	Pass
MULT02	Multiplication with Parentheses	$((2)*(2*4))*10$	160	160	Pass

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

MULT03	Multiplication with unary operators	$+4*3*(-5)$	-60	-60	Pass
DIV01	Division	$5/2$	2.5	2.5	Pass
DIV02	Division with Parentheses	$9/(2/4)$	18	18	Pass
DIV03	Multiplication with unary operators	$+10/5/-2$	-1	-1	Pass
MOD01	Modulo operation	$5\%8$	5	5	Pass
MOD02	Modulo with Parentheses	$(12\%8)\%10$	4	4	Pass
MOD03	Modulo with unary operators	$+9\%-7$	-5	-5	Pass
EXP01	Exponentiation	$3 ** 2$	9	9	Pass
EXP02	Exponentiation with Parentheses	$2**(2--1)$	8	8	Pass
EXP03	Unary Negation and Exponentiation	$+2 ** (-3)$	0.125	0.125	Pass
UNA01	Multiple negative Unary Operators	$--(-2)$	-2	-2	Pass
UNA02	Unary Plus	$++5$	5	5	Pass
UNA03	Multiple Unary Operators	$--+(10)$	10	10	Pass
UNA04	Positive Unary on Negative	$+(-2)$	-2	-2	Pass
UNA05	Negative zero	-0	0	0	Pass

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

MIX01	Mixed Operator	$4 * (3 + 2) \% 7 - 1$	5	5	Pass
MIX02	Mixed Operators with Extraneous Parentheses	$((5 * 2) - ((3 / 1) + ((4 \% 3))))$	6	6	Pass
MIX03	Nested Parentheses with Exponents	$((2 ** (1 + 1)) + ((3 - 1) ** 2)) / ((4 / 2) \% 3)$	4	4	Pass
MIX04	Combination of Extraneous and Necessary Parentheses	$(((((5 - 3))) * (((2 + 1))) + ((2 * 3))))$	12	12	Pass
MIX05	Extraneous Parentheses with Division	$((9 + 6)) / ((3 * 1) / (((2 + 2))) - 1)$	-60	-60	Pass
MIX06	Combining Unary Operators with Arithmetic Operations	$+(-2) * (-3) - ((-4) / (+5))$	6.8	6.8	Pass
MIX15	Combining Unary Operators with Parentheses	$- (+2) * (+3) - (-4) / (-5)$	-6.8	-6.8	Pass
MIX16	Deeply Nested Parentheses	$((((((((((1))))))))))$	1	1	Pass
FLOAT01	Float number expressions	$1.5 + 2.5$	4	4	Pass
FLOAT02	Floating-point number division	$20 / 0.5$	40	40	Pass
FLOAT03	Floating-point number multiplication	$100.5 * 2$	201	201	Pass

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

ERR01	Unmatched Parentheses	$2 * (4 + 3 - 1$	Error: Unmatched parentheses.	Error: Unmatched parentheses.	Pass
ERR02	Operators Without Operands	$* 5 + 2$	Error: Expression cannot start with operator.	Error: Expression cannot start with operator.	Pass
ERR03	Incorrect Operator Usage	$4 / 0$	Error: Division by zero.	Error: Division by zero.	Pass
ERR04	Missing Operator	$5 (2 + 3)$	Error: Missing operator before parenthesis.	Error: Missing operator before parenthesis.	Pass
ERR05	Invalid Characters	$7 \& 3$	Error: Invalid character found.	Error: Invalid character found.	Pass
ERR06	Mismatched Parentheses	$((3 + 4) - 2) + (1$	Error: Mismatched parentheses.	Error: Mismatched parentheses.	Pass
ERR07	Invalid Operator Usage	$((5 + 2) / (3 * 0))$	Error: Division by zero.	Error: Division by zero.	Pass
ERR08	Invalid Operator Sequence	$((2 -) 1 + 3)$	Error: Operator cannot be followed by closing parenthesis.	Error: Operator cannot be followed by closing parenthesis.	Pass
ERR09	Missing Operand	$((4 * 2) + (-))$	Error: Operator cannot be followed by closing parenthesis.	Error: Operator cannot be followed by closing parenthesis.	Pass
ERR10	Invalid Characters	$((7 * 3) ^ 2)$	Error: Invalid character found.	Error: Invalid character found.	Pass
ERR11	Incorrect Operator Usage (modulo)	$7\%0$	Error: Modulo by zero.	Error: Modulo by zero.	Pass
ERR12	Improper float values (more than one decimal point)	$12.3.5 + 4$	Error: Invalid float value.	Error: Invalid float value.	Pass

The Calculatorinator	Version: 1.2
Test Case	Date: 07/05/26
05-Test-Cases	

ERR13	Empty Parentheses	()	Error: Empty parentheses.	Error: Empty parentheses.	Pass
ERR14	Sequential operators	5 + * 2	Error: Invalid operator sequence.	Error: Invalid operator sequence.	Pass
ERR15	Expression ending with operator	5 + 2 +	Error: Expression cannot end with operator.	Error: Expression cannot end with operator.	Pass
ERR16	Improper float values (incomplete float number)	5. + 2	Error: Invalid float value.	Error: Invalid float value.	Pass
ERR17	Improper float values (incomplete float number 2)	.5 + 2	Error: Invalid float value.	Error: Invalid float value.	Pass
ERR18	Divide by 0	7 / 0	Error: Division by zero.	Error: Division by zero.	Pass
ERR19	Empty input		Error: Input cannot be empty.	Error: Input cannot be empty.	Pass
ERR20	Unmatched Parentheses	)3+4(	Error: Unmatched parentheses.	Error: Unmatched parentheses.	Pass